“Explaining AI Medical Models in My Way”: an LLM-enhanced Personalized Report for Clinicians

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***Abstract*—Clinicians are increasingly focusing on Explainable AI (XAI) to better understand AI medical models and improve decision-making. While algorithm-focused approaches dominate, the role of user characteristics—crucial for how explanations are perceived—is critical but often neglected. To address clinicians' diverse explainability needs, we developed a large language model (LLM) powered system that "translates" medical AI outputs into adaptive, characteristics-driven reports. Through a formative study with Parkinson's Disease (PD) clinicians (N=8) and prior studies, we identified that clinicians' preferences for reports, containing information Type, content Tone, and reading Time (3T), vary based on their expertise and workflow urgency. Leveraging these insights, we developed PD Report, an adaptive system powered by large language models (LLMs) to generate personalized, context-aware diagnostic explanations tailored to clinicians' expertise and workflows. A user study (N=12) validated its usability, with clinicians reporting high satisfaction, trust, and efficiency. The system achieved a System Usability Scale (SUS) score of 82.9 (excellent usability), a strong intention-to-use rating (4.8/5), and low cognitive load (NASA Task Load Index). This work highlights LLMs' potential to deliver personalized, understandable outputs from AI medical models.**

***Index Terms*—Human computer interaction, Explainable AI, Smart healthcare, User experience, Human-centered computing.**

I. Introduction

Artificial Intelligence (AI) technologies, such as AI-Clinical Decision Support Systems (CDSSs) [1], are increasingly developed in healthcare to improve clinical efficiency. However, rising concerns of transparency [2] and accountability [3] highlight the need for Explainable AI (XAI) to build trust and support informed decision-making [4] especially in high-stakes domains. While most existing works focus on algorithm-centric approaches to enhance model transparency, it is often neglected to consider who will use these models and how models will be used [5]. In contrast, the Human-Computer Interaction (HCI) community emphasizes a human-centric perspective, recognizing that users' characteristics fundamentally shape how explanations are perceived and understood [6]. Prior studies have explored healthcare users' different preferences for explanation tone [7], information type [8], amount of explanations [9], and when and where to provide explanations [10], offering valuable insights. However, a gap remains in integrating these diverse needs into a single system that delivers personalized, workflow-aligned explanations for clinicians.

Motivated by this, we focused on "translating" a medical AI for Parkinson's Disease (PD), where the wearable IMU data was leveraged as the input and a score of Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [11] was predicted. PD, a progressive neurodegenerative disorder [12], is diagnosed primarily through subjective evaluation of motor symptoms, often leading to high inter-rater variability due to differences in clinicians' experience and patients' self-reports [13]. Thus, an objective assessment CDSS of PD can greatly enhance clinical quality, assist less experienced clinicians, and save time. However, two challenges hinder effective implementation: (1) compared to medical images, wearable data outputs are more challenging to interpret and relate to patient actual conditions [14], requiring explanations aligned with clinicians' mental models, and (2) clinicians' diverse backgrounds and experiences [15] necessitate personalized reports to cater to individual needs.

To understand the relationship between clinician characteristics and report preferences, we reviewed prior studies and conducted a formative study involving eight PD clinicians with varying expertise and hospital backgrounds. Results revealed that clinicians' preferences for report content were shaped by two key factors: expertise and urgency. Experienced clinicians prefer concise, actionable summaries focusing on efficiency, while less experienced ones seek detailed explanations for learning and decision-making, aligning with prior findings [7]. In urgent scenarios, clinicians valued quick-to-read formats and key explanation information, whereas less time-sensitive cases allowed for more comprehensive reports. Across the board, clinicians favored visualized results and natural language explanations, consistent with their habits of reviewing medical case notes. In summary, clinicians' preferences were categorized with "3T" factors: information Type, content Tone, and reading Time, influenced by their expertise and urgency.

Inspired by these findings, we developed PD Report (Figure 1), a novel system powered by a large language model (i.e., DeepSeek-V3) to generate a dynamic and personalized report for explaining AI models. LLMs have proven to be effective, low-develop-cost tools for generative, multi-purpose agent collaboration, which have demonstrated their potential in CDSS

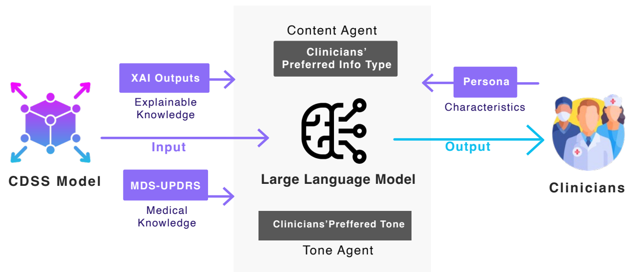

Fig. 1. Overview of PD Report

applications by recent works [16], [17]. The system built three AI agents: (1) **Content Agent**, determining information types based on expertise and urgency (via few-shot learning); (2) **Tone Agent**, determining the tone based on expertise (via few-shot learning); (3) **Report Agent**, generating tailored reports based on outputs of AI and XAI models, medical knowledge (via Retrieval-Augmented Generation, RAG), and preferred format and reading time. This approach bridges the gap between AI outputs and clinicians' mental models, presenting an explanation report that aligns with their expectations.

We evaluated the PD Report with 12 clinicians from varied backgrounds. The system achieved a System Usability Scale (SUS) score of 82.9, reflecting excellent usability, and reported low workload based on the NASA Task Load Index (NASA-TLX). Clinicians rated their willingness to adopt the system at 4.8/5, efficiency at 4.4/5, and its potential to replace traditional at-home assessments at 4.5/5. Notably, 11 out of 12 participants favored PD Report over a comparison version created without LLMs. Interviews confirmed the system met clinicians' expectations and provided actionable insights. Moreover, we discovered some unexpected effects of LLM employment.

II. Background and Related Works

A. AI-based CDSS for PD

PD is a progressive neurodegenerative disorder [12], with clinical diagnosis relying on MDS-UPDRS [11], which evaluates motor symptoms (e.g., gait, tremor, bradykinesia [18]) and cognitive symptoms [19], [20]. To address the subjective variability of clinician experience [13], [21], objective AI-CDSS systems (e.g., Hopkins PD [22], M Power [23], Roche PD [24]) have emerged. These systems analyzed signal inputs to make diagnostic predictions, but presenting results in a clinician-friendly, interpretable manner remains challenging.

B. Human-centered XAI in healthcare

Unlike algorithm-focused XAI, human-centered XAI in healthcare emphasizes usability and interpretability [25]. Most research in this area focused on fostering appropriate trust [26] and designing systems for real clinical settings [10], [27]. Calisto et al. [7] explored how clinicians' professional experience influences their preferred communication tone of AI results; Qian et al. [9] proposed "Unremarkable AI," delivering AI results only when risky to high-urgency clinicians; and Liang et al. [28] emphasized tailoring explanations to users' knowledge backgrounds.

While these studies provided valuable insights and solutions, few addressed the challenge of delivering personalized explanations within a single system. A recent review [29] found that healthcare stakeholders' explainability needs were related to expertise and urgency. Thus, clinicians' needs vary significantly in real-world practices, making one-size-fits-all explanations insufficient [30]. Additionally, compared to medical images that clinicians can read directly, wearable device-based AI-CDSS presents more challenges for understandable outputs.

C. LLMs for Healthcare

LLMs have gained attention for their ability to generate diverse, personalized content. Beyond ChatGPT [31], models like Deepseek [32] gained attention recently by scalability and faster inference speed, particularly for non-English content. LLMs are increasingly adopted for healthcare applications, including mental health support [33], self-diagnosis [34], and healthcare communication [35], among others. Notably, Rajashekar et al. [16] explored LLMs in AI-CDSS, demonstrating their potential to enhance usability and trust. By integrating knowledge from external databases through RAG [36], LLMs can be further strengthened for knowledge-intensive healthcare domains. These advancements highlight LLMs' ability to deliver personalized knowledge, inspiring our approach. However, there persist ethical concerns, such as bias, inconsistency and potential errors [37], [38], which were considered in our development.

III. Formative Study

To understand user needs and explore the relationship between clinician characteristics and their preferred explanation styles, we conducted semi-structured interviews aimed at discovering (1) user needs, (2) preferences for explanation presentation, and (3) potential concerns.

A. Participants and Procedure

We recruited eight clinicians with diverse backgrounds: four from a top-tier hospital with a dedicated PD department (indicating high expertise), and four from non-top-tier hospitals without a PD department but with neurology units. Their professional experience ranged from 2 to 31 years, and none had prior experience with the PD CDSS application. Detailed demographics are in the Supplementary Materials.

The semi-structured interview focused on three main goals: (1) understanding their workflows and diagnostic priorities in PD, (2) identifying how they preferred results to be presented and whether this correlated with their characteristics, and (3) discovering concerns about system adoption.

We used a pre-interview questionnaire to understand their workflows, demonstrated how patients use the PD CDSS application at home, and presented six pre-prepared XAI mockups: key features, mean values, Pearson's Coefficient, LIME, visualizations (original, similar, and counterfactual),

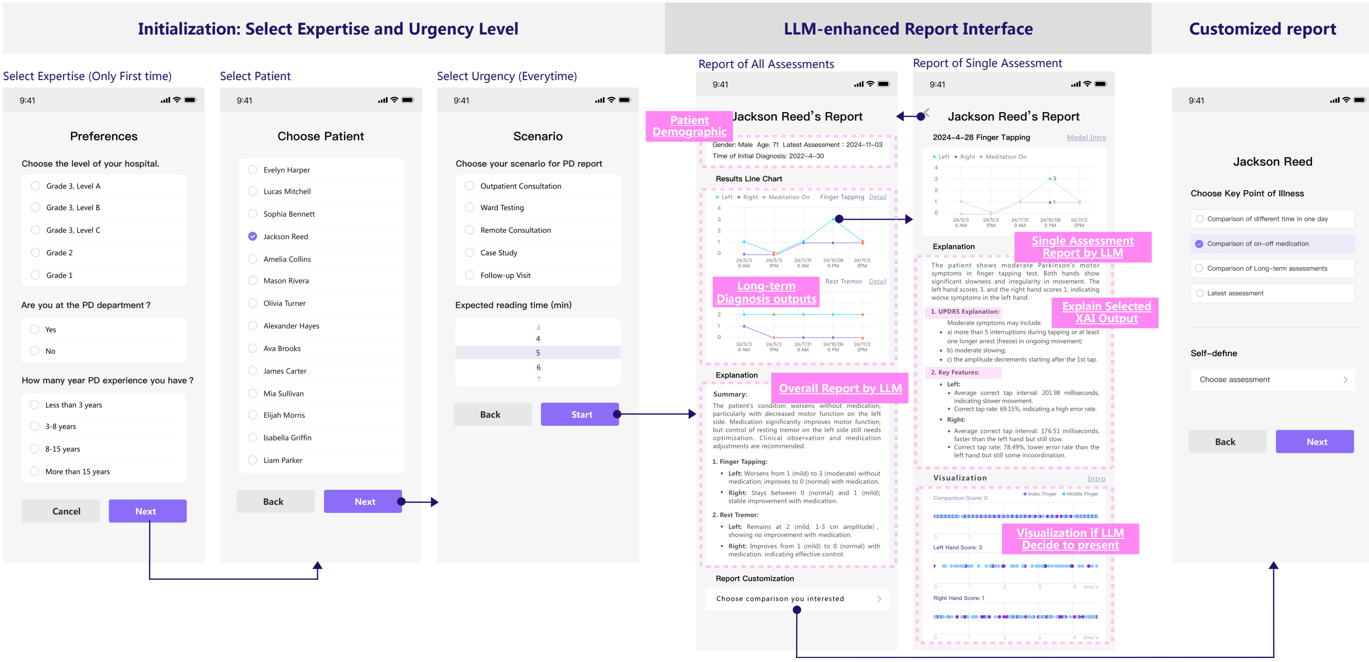

Fig. 2. Prototype of PD Report, translated to English from the original language.

and MDS-UPDRS explanations. These mockups focused on common PD assessments, Finger Tapping and Rest Tremor. Follow-up questions addressed their responses to the three goals. Each interview lasted about 1.5 hours, and participants received a $30 gift card. Interviews were conducted on-site and recorded using a GoPro.

B. Findings

1) User Needs and Challenges in PD Diagnosis: Clinicians described their challenges with PD traditional diagnosis and outlined areas where AI could assist. Key findings were:

Subjectivity and time constraints are major challenges. Six clinicians highlighted subjectivity as a primary issue, while four cited time constraints. For instance, P7 mentioned inconsistencies in UPDRS ratings even among experienced clinicians. They noted that while minor rating gaps are acceptable, AI's consistency could provide consistent value, especially in tracking patient progress over time. All participants expressed openness to adopting AI tools, with P2 stating, "I've always wanted such a system in clinical settings. If usability is validated, our hospital would consider implementing it."

Scenarios where CDSS could add value. Clinicians identified scenarios where AI-CDSS could enhance their work, such as reexaminations, follow-ups, remote consultations, and at-home medication adjustments. These scenarios often lack objective data, making AI-CDSS a valuable supplement to subjective patient reports. Clinicians also emphasized the importance of core assessments like Finger Tapping and Tremor in time-constrained outpatient settings.

Shared preferences despite diverse needs. Clinicians preferred simple, effective results especially for time-sensitive scenarios (e.g., outpatient visits). P4 remarked, "I'd lose patience if I can't get it in 5 seconds." At the same time, most clinicians (6 out of 8) preferred explanations in natural language, aligning with their experience reviewing patient case notes. Common focal points for illness included long-term trends, on-off medication effects, and intra-day variations.

2) Relationship between Characteristics and Preferences: Clinicians' needs varied but followed patterns based on expertise and urgency, which we categorized as 3T preferences: information Type, content Tone, and reading Time.

Expertise. Expertise, determined by hospital level and years of PD experience, influenced preferences for explanation types and tone:

High-expertise clinicians sought global explanations (e.g., model performance) to validate system accuracy. P8 (19 years experience) stated, "I need to check the model's performance to decide if I can trust it." They also showed interest in how the model works, likely due to their research experience.

In contrast, less experienced clinicians viewed the system as a learning tool, preferring local XAI results (e.g., MDS-UPDRS explanations, visualizations) to support their understanding. P2 noted, "These results can help young clinicians by providing references when they lack confidence." Thus, it is crucial to generate appropriate trust for novice to match their limited confidence. This aligns with prior findings that low-expertise clinicians prefer assertive tones, while high-expertise clinicians favor suggestive tones [7].

Urgency. Scenario urgency influenced reading time and information preferences: In high-urgency scenarios (e.g., outpatient visits), clinicians expected to spend 1–2 minutes reviewing reports and focused on one or two key outputs. In low-urgency scenarios (e.g., follow-ups), they preferred more detailed information. P1 explained, "I don't have time to check

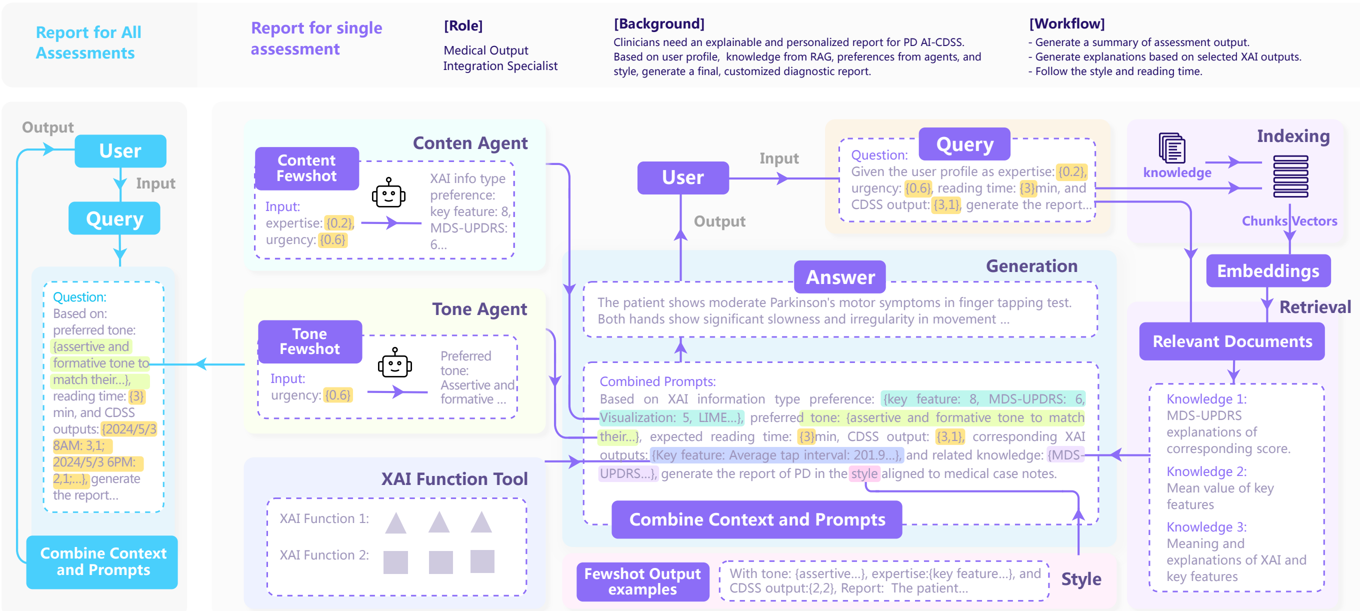

Fig. 3. Pipeline of LLM generation in PD Report

all XAI information during outpatient visits, but I'd review it in seminars for complex cases."

3) Concerns: Clinicians with research experience raised concerns about LLM reliability and inconsistency as we had. Since it's beyond our focus and too big to tackle or ignore now, we mitigated this issue by inviting a PD clinician to review all LLM-generated reports post-evaluation, ensuring only error-free outputs were included.

4) Findings Summary: 1) Clinicians faced subjective and time-consuming PD diagnosis challenges and were eager to adopt AI tools.

2) Explainability needs varied based on expertise and scenario urgency, forming the basis for our personalized report design.

3) Except for their different needs, they shared preferences for the format of the report (visualized results, clear natural language explanations), scenarios for CDSS Empowerment, and Common focal points for illness.

IV. Design and Implementation of PD Report

Delighted by insights from formative study, we designed an LLM-powered PD Report to generate personalized explanations for AI-CDSS.

A. Design

Figure 2 outlines the interaction flow for the PD Report, designed from the end-user's perspective. It consists of two main phases: Initialization and Report Generation.

1) Initialization: The first step involves gathering the clinician's expertise and scenario urgency. Expertise is a one-time setup for first-time users, as it's a stable characteristic determined by three questions: hospital level, PD department status, and years of PD experience. Non-first-time users only need to select a patient and the scenario's urgency, which adjusts the expected reading time automatically (modifiable if needed). By default, five urgency levels corresponding to typical scenarios identified in the formative study are provided to minimize interactions and enhance efficiency.

2) Report Generation: The report page integrates three main components:

1) Line Chart for trend analysis: displays up to five recent assessment results, showing long-term trends. Clicking a specific result navigates to a detailed single-assessment report.

2) LLM-Generated Explanations: Offers concise, efficient, and personalized insights aligned with clinicians' reading habits. (pipeline listed in Figure 3)

3) Customized comparison: allows customized assessment comparisons and generate reports again to meet specific diagnostic needs.

The default page also includes patient demographic data for context. The main report highlights overall trends and potential warning signs, while single-assessment reports delve deeper, featuring explainable content such as LIME, MDS-UPDRS, key features, and more. Personalized reports leverage preferences set during initialization, with adjustments for tone, length, and XAI methods. Visualizations are added when deemed appropriate by the LLM.

B. Implementation

The PD Report is an enhanced version of the original PD AI-CDSS Android app, incorporating personalized features. The implementation consists of Client and Server-Side Functions and LLM Integration.

1) Client and Sever-side Implementation: The client side is an Android app that implements all the features of our design, as mentioned above. User inputs (expertise and urgency) are obtained through the API. After initialization, the client side sends collected data to the server side for processing. The server side generates personalized reports based on assessment data, but creating a single multi-agent report takes over 30 seconds, which can frustrate users. To address this, the server pre-generates all 10 possible individual reports immediately after initialization. Additionally, the server supports LLM-based generation as well, including invoking the Deepseek API, retrieving from a knowledge database, and coordinating multiple agents.

2) LLMs Implementation: Due to the faster inference speed and better performance of generating non-English language, we choose Deepseek-V3 as our LLM . Three agents collaborate to deliver personalized reports to achieve the goal of personalized reports with the clinician's preferred information type, content tone, and reading time (shown in Figure 3).

Content agent: Determines XAI method preferences (key features, LIME, visualizations, etc.) based on expertise (E) and urgency level (U). E and U are defined as a value between 0–1. Preferences were collected from 8 clinicians across outpatient and follow-up scenarios (high vs. low urgency). Few-shot learning was used to enhance understanding of limited data, with E calculated as $E = H * D * Y$, where H, D, and Y represent hospital level, PD department status, and years of experience, respectively. H Y are normalized, and D (if PD department) categorized into $\{0.5, 1\}$.

Tone agent: It is similar to Content agent, facilitating by few-shot prompting. Adjusts the content tone (suggestive vs. assertive) based on E. Higher expertise correlates with suggestive tones, while lower expertise prefers assertive tones.

Report agent: The report agent consolidates all available information into final prompts and produces the ultimate output. Apart from considering content and tone, the agent also focuses on optimizing reading time and format to align with clinicians' preferences. Additionally, the LLM agent accesses XAI information by utilizing XAI functions and retrieving relevant knowledge from a database via RAG [36]. Typical XAI functions LIME, key features and visualizations (original, similar, and counterfactual) compromised a function pool for the agents as tools. The knowledge database encompasses diverse aspects of application knowledge, including the physical significance of extracted features, explanations of MDS-UPDRS, average feature values, Pearson's Coefficient, and more. In essence, the Report agent combines preferences, domain and model knowledge, as well as CDSS model and XAI model outcomes as prompts to generate adaptable, individualized, and comprehensible final reports.

V. Evaluation

We conducted a user study to evaluate PD Report, assessing whether it meets clinicians' personalized needs. Results showed that the system provides a helpful, effective, and comprehensible experience for PD diagnosis. We also uncovered several unexpected insights.

A. Participants and Procedure

We recruited 12 clinicians with diverse backgrounds: six from a top-tier PD department (including three undergoing in-service training) and six from a hospital without a PD department but with a neurology department. All participants possessed experience in PD diagnosis, ranging from 3 to 32 years. None had prior exposure to our PD-CDSS tool. Detailed demographic information is available in supplementary materials.

We conducted a user study on-site, and they got a gift card of about 30 dollars after finishing the interview, which lasted for about one hour. The study is made up of an introduction, pre-interview, scenario task, and semi-structured interview about their test experience. In the introduction, we briefly introduced our system and asked them to read and sign the informed consent if they had no questions. We asked for their demographic information and impression of AI-CDSS before they used PD Report in the pre-interview. To evaluate the performance of PD Report in different scenarios, participants were asked to imagine themselves in outpatient and remote consultation scenarios to evaluate the PD Report's performance. We presented two sets of patient data with fake names to protect their privacy, including detailed assessments from our PD CDSS application. Participants were exposed to both the PD Report and a comparison report lacking LLM-powered explanations to gauge their preferences. To maintain rigor, half the participants viewed the PD Report first, while the other half started with the traditional report. Following the tasks, follow-up interviews were conducted to gather feedback, with a pre-designed questionnaire provided in the supplementary materials. Notably, we invited a PD clinician to review all LLM-generated reports post-evaluation, ensuring only error-free outputs were included.

Also, two questionnaire tools are included in the interview. NASA Task Load Index (NASA- TLX) [39] on a seven-point semantic differential scale to assess participants' subjective workload in professional and complex tasks, including Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. A lower score indicated a lighter workload and a better outcome in TLX. The System Usability Scale (SUS) [40] on a five-point Likert scale to assess the usability of PD Report. In half of 10 questions, a higher score means better usability, while others are the opposite.

B. Usability

1) High Usability in Understanding, Helpfulness, and Efficiency. Participants expressed positive feedback, highlighting its potential for future implementation in clinical settings (results provided in the supplementary materials). It achieved a SUS score of 82.9$\pm$7.3, indicating excellent usability. Helpfulness was rated 4.3$\pm$0.5, and all participants expressed a strong intention to use the system (4.8$\pm$0.5). all of them said they would like to use it. For example, PP7, as a fresh clinician, noted, "If more assessments were included, this could replace some of our diagnostic work in the ward. It

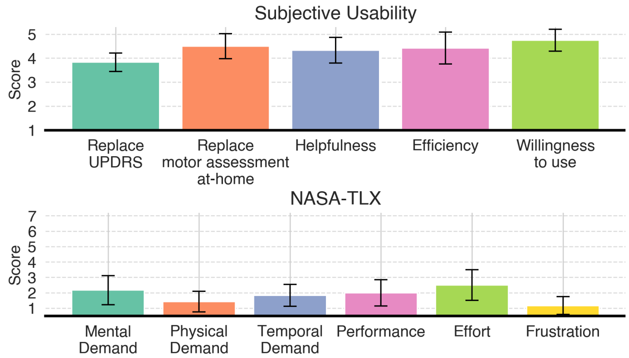

Fig. 4. Subjective usability and NASA-TLX (workload) rating of PD Report.

simplifies examinations, and I think our hospital would adopt it, even with payment."

Regarding the potential replacement of MDS-UPDRS (3.8±0.2) by the system, in contrast to the option of substituting motor assessments conducted at home (4.5±0.4), most of the participants were concerned if PD Report can replace the whole traditional assessments due to they did not trust AI-CDSS for 100 percent, double check is still needed. However, they highly agreed to use it to assess patients at home. PP3 said "It's obviously a good idea to employ it in scenarios where we cannot assess patients ourselves, I mean, when they are at home."

Participants also endorsed the effectiveness (4.4±0.7) of PD Report. PP11 mentioned "I cannot imagine how much time would be saved if patients come with this report. It'll take several minutes to figure out what happened to them just by talking cause most of the illness change occurred during the past years." PP5 also said, "It's very effective, only need to click several times and then show the results." Participants built appropriate trust as well as all participants mentioned they wouldn't trust the report 100 percent, but to validate in their own ways. PP2 explained "I trust the result in about 80%, but need to test the result with my assessment to figure out if it is really trustworthy, and I guess only a few times validations are enough." PP9 said that "It's okay if the AI result is different from mine, only if it is consistent to itself, but it is model, I think it's not hard for it, right?"

2) PD Report Outperforms Non-Personalized Version. Eleven of 12 participants preferred PD Report over the non-personalized version. They found the comparison version too lengthy for time-sensitive scenarios (e.g., outpatient visits) and lacking explanations, making it harder to understand. One expert (PP9) disliked its uniform tone, highlighting the benefits of LLM-powered personalization.

3) Low workload. The NASA-TLX score as listed in Figure 4 indicated low workload across all dimensions: mental demand (2.2±0.9 out of 7), physical demand (1.4±0.7), temporal demand (1.8±0.7), performance (2.0±0.9), effort (2.5±1.0), and frustration (1.2±0.6). Slightly higher effort scores likely reflect the learning curve for a new system, but overall workload remained low.

C. Findings

1) Experts question AI, novices question themselves. The reactions of clinicians to differing decisions between themselves and AI depend on their expertise levels. Experts tend to question the AI system when their decisions conflict, as they have high confidence in their abilities, seeing it as an opportunity to validate the AI's reliability. Novices, on the other hand, are more inclined to doubt their own judgment and seek to understand the AI's rationale. Novices reported lower effort and performance in self-assessment compared to experts, indicating greater difficulty with diagnosis tasks. Tailoring communication tones is crucial, with novices benefiting from a supportive, assertive approach akin to guidance from a supervisor, while experts prefer a more suggestive tone.

2) LLMs generation matters more to novices. Our research revealed that clinicians with lower expertise levels tend to refer to LLMs reports more frequently than those with higher expertise. Novices found these explanations helpful in comprehending patient cases better. For instance, PP12 stated, "I appreciate these explanations as I'm not always familiar with UPDRS. They help me in understanding the case and learning simultaneously." PP4 said "Only a line chart is definitely not enough for me. I need to check the explanation to either verify my diagnosis or learn from it. Maybe it's okay for my director, but not me." In contrast, experts show less reliance on explanations, often focusing on the line chart results for quick verification. When asked if the LLM content is not useful for them, they said it's necessary to have them there in case they need, or sometimes as a support for their uncertain decision. PP9 said that "No no no, definitely keep it. The content here is pretty good and helpful. Just the cases you provide here are clear, I can see what happened to their situation just based on the line chart results."

3) LLMs Build Trust and Support Coaching. Participants expressed that the LLMs reports enhanced their understanding of patient's subtype, for example in Finger Tapping assessment, key feature and its explanations helped participants know the patient is bad because of low tapping speed, or irregular tapping. PP8 said "This is effective to know his detailed case, and knowing this is important for his treatment in the following." Except this, LLMs show effects beyond our thoughts. First, some of the high-expertise participants use it to build trust. PP9 checked line chart results and mumbled her inference, then found similar content when she read the LLM's explanation. She said "See? The same opinion." She expressed she built trust mainly from that moment. Second, it might help experts to coach their students. PP11, with 18 years of experience, said that "This explanation is really good to teach my students how to write case notes." She thought the LLM results were standard, and it would also be great if they could be copied to her case notes automatically.

VI. Discussion

The user study confirmed the usability of personalized PD Reports for diverse users in various scenarios. This section delves into the potential of applying LLM-enhanced AI-CDSS

Report pipeline in other cases in the healthcare domain, existing constraints of LLM-enhanced AI-CDSS, the importance of appropriate trust and ethical concerns, as well as Limitations and Future Work.

A. LLM-enhanced AI-CDSS Report

Prior works have shown the empowering potential of LLMs for CDSS [16]. Our studies align clinicians' mental models with AI, showcasing LLMs' ability to deliver tailored reports. Clinicians, depending on their expertise, require varying levels of detail in explanations. LLMs can adapt to these needs through few-shot learning, offering detailed references for novices and concise summaries for experts. Additionally, integrating domain knowledge via RAG supports the explanation of AI outputs, aligning with clinical practices. Also, natural language can align with clinicians' habits of reading case notes and adjusting the tone to meet their preferences, and then increase trust. The adaptability of report generation to clinical urgency further integrates AI into medical workflows, suggesting the pipeline's applicability to other medical models for producing clinician-friendly XAI outcomes.

B. Constrains of LLM-enhanced AI-CDSS

Despite the advantages, LLM-enhanced AI-CDSS faces challenges. Compared to pre-defined reports, LLM-generated results inevitably take more time. In our work, to mitigate the extended time caused by multi-agent systems, we pre-generated results, but this still couldn't fully address all user experiences. For instance, two users who quickly clicked on results which still being generated showed significant impatience, which also led to reduced engagement with single assessment pages in future use. Moreover, structural consistency is maintained through temperature settings and few-shot learning, yet balancing the trade-off between consistency and creativity remains a hurdle, potentially limiting LLMs' effectiveness in certain applications.

C. Appropriate Trust and Ethical Concerns

As observed in the user study, experts are highly confident in their own judgments, while novices tend to lack confidence, leading to issues with appropriate trust. While an appropriate content tone can help to some extent, its impact is not enough. Appropriate trust is always a tricky problem that appeals to many HCI researchers to devote [26]. It's crucial to ensure human involvement in CDSS, as humans are ultimately the responsible agents. AI can make mistakes, and over-reliance is far more problematic than a lack of trust. This highlights an area for further optimization of the pipeline through natural language or interactive systems to keep doctors engaged.

D. Limitations and Future Work

Our work has certain limitations and corresponding future directions. First, we originally designed an optimization mechanism where clinicians could review LLM-enhanced reports and provide feedback (e.g., thumbs up or down) to refine the prompts and examples, enabling a human-in-the-loop process to improve results. However, since this study is one-round, implementing and validating such feedback requires significant time, a huge amount of expertise, and corpus accumulation. As a result, we omitted this part, though the pipeline is technically capable of supporting this functionality. We hope to further develop this in the future, allowing the system to adapt and continuously improve based on user feedback.

Second, while the PD AI-CDSS is primarily designed to assist clinicians, it also has the potential to address the needs of other stakeholders, such as self-monitoring patients, their family members, or caregivers. Providing personalized explanations tailored to different stakeholders' purposes can also be achieved using the current pipeline. This direction holds great promise for building a multi-stakeholder CDSS, and it will be a focus of our future work.

Finally, this study lacks certain quantitative results. Due to its emphasis on personalized reports, the evaluation mainly relied on subjective feedback from users. In the future, we aim to incorporate additional objective methods, such as timing-based metrics, to assess efficiency improvements and enhance the study's objectivity.

VII. Conclusion

In this paper, we designed and implemented PD Report to generate personalized CDSS reports for different clinicians. Initially, we conducted a study (N=8) to examine how clinician attributes relate to their preferred explanation styles. Our results showed that clinicians' choices of XAI information, content tone, and reading time (3T) are influenced by their expertise and workflow urgency. Guided by insights, we developed an LLM-enhanced system to generate tailored explanations for PD diagnosis, considering clinicians' expertise and workflow. Through a user study (N=12), we assessed the system's usability based on helpfulness (4.3 out of 5), efficiency (4.4), potential to replace assessments (4.5), and intention to use (4.8). An overall SUS score of 82.9 indicated excellent usability, and NASA-TLX scores confirmed low cognitive load. This research delves into LLMs' capability to create personalized and clear outputs from AI medical models.

VIII. Acknowledgment

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